MIL-A-21577/8A(SHIPS)
27 February 1967

SUPERSEDING
MIL-A-21577/8(SHIPS)
15 October 1959
(See 6.2)

MILITARY SPECIFICATION

# AMPLIFIER-CONTROL GROUP, AN/SIA-120

## 1. SCOPE

1.1 This specification covers the detailed requirements for an audio frequency amplifier control group type AN/SIA-120.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL - C-62 - Capacitors, Fixed, Electrolytic (DC Aluminum, Dry Electrolyte, Polarized) General Specification for.
MIL-C-3965 - Capacitors, Fixed, Nonsolid Electrolytic, (Tantalum, Foil and Sintered Slug) General Specification for.
MIL-A-21577 - Amplifiers, Audio Frequency; and Amplifier Control Groups—Shipboard Announcing.

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance Lists for Semiconductor Devices.

DRAWINGS

BUREAU OF SHIPS

9000-S6502-74117 - Receptacle, (MIL-R-2726) Announcing Equipment Plug and Receptacle.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Amplifier Control Group AN/SIA-120 shall be in accordance with all applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 The equipment complement shall consist of two power amplifiers and all relays, meters, switches, terminals and wiring required to provide control switching and amplification of voice signals for use in announcing system, circuit 2MC, mounted in a common bulkhead enclosure.

3.2.1 Preamplifiers shall be included as a part of each power amplifier.

3.3 Power amplifiers shall be modular construction with quick disconnect features to permit easy removal of the amplifiers for servicing.



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3.4 Capacitors. - Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in amplifier assemblies. Electrolytic capacitors in accordance with MIL-C-62 may be used in power supply filter circuits only.

3.5 Semiconductors. - Semiconductor circuits shall be used exclusively. Semiconductors shall be approved types listed in MIL-STD-701.

3.6 Relays with the exception of power type contactors may be hermetically sealed.

3.7 Printed wiring shall be in accordance with stand off type lead terminations specified in MIL-STD-275.

3.8 Output power. - The output power of the individual power amplifier units shall be 20 watts minimum rating.

3.9 Input facilities. - Input facilities shall be provided to accommodate five microphones (type IC/MPM or type IC/MFM). A test input receptacle assembly (Drawing 9000-S6502-74117) to permit connection of a microphone shall be provided on the AN/SIA-120 control rack front panel. In addition, wiring shall be provided to the terminal block, from the amplifier terminals to provide insertion of alarm signals from an external source.

3.9.1 Each of the five microphone circuits, shall be independently connected by means of relays to the amplifier input so that only the microphone in use is connected to the input of the amplifier.

3.9.2 The two amplifier inputs shall be connected in parallel and connected to the output of the microphone relays through a suitable resistance network to provide the required 150 ohms input impedance.

3.10 Five loudspeaker output groups shall be provided. Three position switches shall permit connecting any or all of the output groups to the output of either of the amplifier assemblies. Parallel connection of the outputs of the two amplifiers will not be required.

3.10.1 A local loudspeaker cutout relay shall be included for each microphone circuit to disconnect local loudspeakers when the microphone is in use.

3.11 Circuit transfer switches. - Circuit transfer switching shall be as specified in 3.9 and 3.10.

3.12 Output metering and switching. - An output volume indicating meter and its associated output selector switch shall be provided, and shall make it possible to measure the output of either amplifier selected for test.

3.13 Maximum weight and size. - The maximum weight and size of the AN/SIA-120 amplifier control group shall be: weight, 100 pounds; height, 20 inches; width, 10 inches; and depth, 10 inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.2 Sampling, examination, and tests shall be as specified in MIL-A-21577.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-21577.



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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging, and level of packing required (see 5.1).

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
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